The bias interval of the elephant random walk

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Abstract

Guérin, Laulin, Raschel, and Simon conjectured that the initial biases giving the superdiffusive elephant random walk a bimodal limit law form an interval centered at one half, and they described the problem as challenging. We prove this. For every superdiffusive memory, strongly biased starts are unimodal, and the limit density has two local maxima exactly when the bias lies in a proper open interval. Outside it, including its endpoints, the density has a unique local maximum. The interval is nonempty exactly above a single memory threshold, numerically about 0.6688, whose uniqueness follows from a positive-weight integral for the central curvature of the fair-start density. We control sign changes in the finite walk's first and second differences, including those created at the boundary, and pass them to the limit. The bias boundary is continuous and tends to zero at least quadratically fast as memory tends to one.

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1 Introduction

An elephant and an ant are playing hide and seek. The elephant counts while the ant hides among many identical anthills, then opens its eyes and knows at once which one the ant is in. The ant's shoes are outside the door.

(Anonymous)

The elephant walk [1] has steps $X_i \in \{+1, -1\}$. Given $X_1, \dots, X_t$, it picks i uniformly from $\{1, \dots, t\}$. Then $X_{t+1} = X_i$ with probability $1 - \varepsilon$, and $X_{t+1} = -X_i$ otherwise. These choices are independent of each other and of X_1. Put

$$A_N = \#\{i \le N : X_i = +1\}, \qquad S_N = 2A_N - N, \qquad P_N(k) = \mathbb{P}(A_N = k).$$

The memory is $a = 1 - 2\varepsilon$ and the initial bias is $q = \mathbb{P}(X_1 = 1)$. In the original description the copying probability is $p = (1 + a)/2$. After t steps with count k, the next step is $+1$

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with probability $r_t(k) = \varepsilon + ak/t$. We call $q = 1/2$ a fair start, and $\mathbb{P}_+$, $\mathbb{E}_+$ condition on $X_1 = 1$.

When $a \in (1/2, 1)$ the ratio S_N/N^a converges almost surely [6, 10]. Let h_a be the density of the limit given $X_1 = 1$. With initial bias q the limit density is

$$g_{a,q}(u) = qh_a(u) + (1-q)h_a(-u), \qquad g_a = g_{a,1/2}.$$

Guérin, Laulin, Raschel, and Simon prove that h_a has one peak [12, Theorem 1.1]. In [12, Remark 2.1(b)], they conjecture that $g_{a,q}$ has 2 peaks exactly for q in an open interval centered at $1/2$, and they describe this problem as challenging. Our Theorem 1.1 answers it with a yes. Furthermore, our Theorem 1.2 states that the interval is empty for $a \le a_*$ and nonempty for $a > a_*$, where $a_* \approx 0.6688$.

1.1 The bias interval and its threshold

Theorem 1.1 (the bias interval)**.** *For every $a \in (1/2, 1)$ there is $b_a \in (0, 1/2]$ such that $g_{a,q}$ has exactly two local maxima if and only if $b_a < q < 1 - b_a$. Outside this open interval, including its endpoints, the density is unimodal with a unique local maximum.*

In particular $g_{a,q}$ is unimodal for $q \le b_a$ and for $q \ge 1 - b_a$. Theorem 1.2 gives the set of a with $b_a < 1/2$.

Theorem 1.2 (the analytic threshold)**.** *There is exactly one $a_* \in (2/3, 69/100)$ with $g''_{a_*}(0) = 0$. The central curvature is negative below a_* and positive above it, and $\partial_a g''_a(0)|_{a=a_*} > 0$. The bias interval is empty exactly when $a \le a_*$.*

Theorem 1.1 is Theorems 3.1 and 3.3 together with $b_a > 0$ from Theorem 3.5. Theorem 1.2 is Theorem 3.2. These proofs use no computer. The bias boundary is regular. By Theorem 3.5, the function $a \mapsto b_a$ is continuous on $(1/2, 1)$, continuously differentiable on $(a_*, 1)$, and strictly decreasing immediately above a_*. By Corollary 3.6, $b_a \to 0$ as $a \uparrow 1$, and by Proposition 3.7, $b_a \le 82(1-a)^2$ for a close to one. We do not know if b_a is decreasing on all of $(a_*, 1)$ (Section 5).

At a_* the fair density has a quartic top (Lemma B.2), and the next theorem describes how the two modes and the bias interval open from it.

Theorem 1.3 (the critical exponents)**.** *There are positive constants K_*, Q_* such that the positive mode κ_a of g_a and the bias boundary satisfy*

$$\kappa_a \sim K_*\sqrt{a - a_*}, \qquad \tfrac{1}{2} - b_a \sim Q_*(a - a_*)^{3/2} \quad (a \downarrow a_*).$$

The two asymptotics are Propositions 3.8 and 3.9. Theorem 3.1 also counts the inflection points of g_a.

Remark 1.4 (numerical values)**.** Numerically, $a_* \approx 0.66881$, $K_* \approx 7.2241$, and $Q_* \approx 9.8670$. These values are not used in any proof. Section 4 describes their interval certificates.

1.2 From finite signs to the limiting shape

Guérin, Laulin, Raschel, and Simon prove unimodality by induction on N for the count law, given $X_1 = 1$. We do the same induction on every q. By (2.2), for $2 \le k \le t-1$ the value $D_{t+1}(k)$ is a positive combination of $D_t(k-1)$ and $D_t(k)$. By Lemma 2.2, this step adds no sign change. The new problem lies at the two ends of a row, because $D_{t+1}(1) = D_{t+1}(t) = s_{t+1}$ is new, and any value added to the end can induce a sign change. Hence, we fix the final length N and write the boundary values $s_N, \dots, s_{t+1}$ entering before time N on both sides of row t. Lemma 2.3 bounds the signs of $(s_3, \dots, s_N)$, and then the induction runs over the whole longer row (Theorem 2.4). For an arbitrary

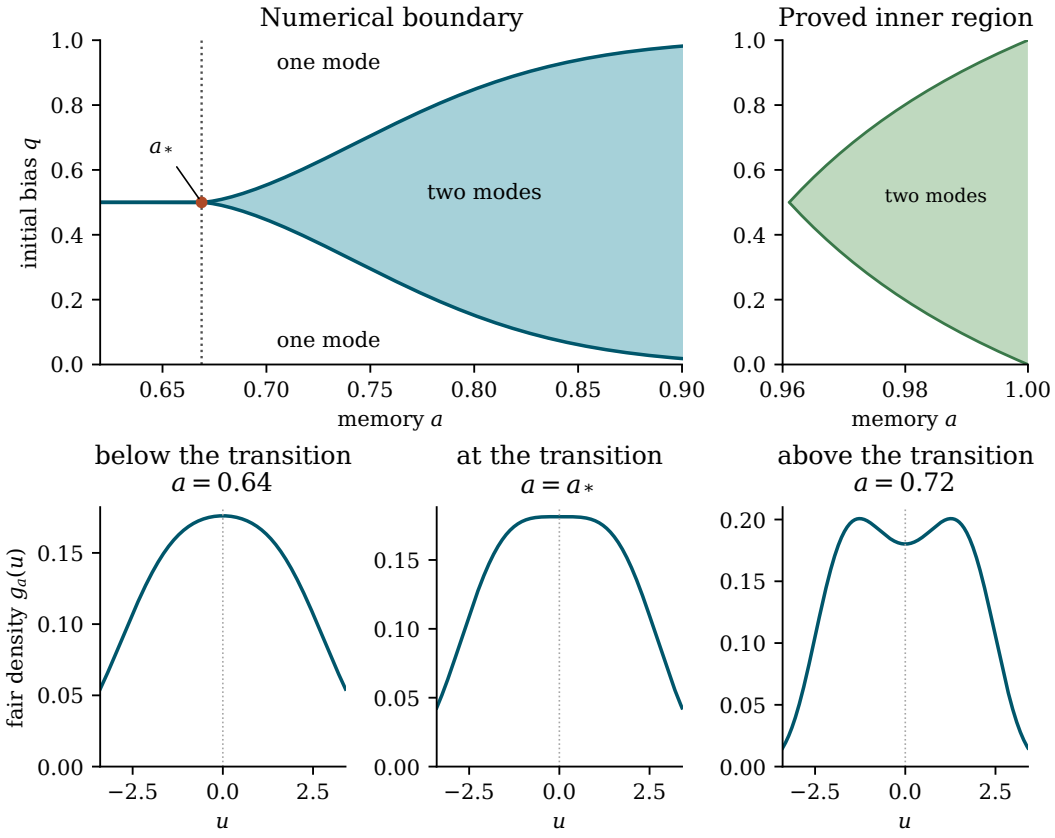


Figure 1: The bias interval and the fair-start transition. Upper left: b_a and $1 - b_a$ from Theorem 3.5, computed numerically (not certified). Upper right: the region $4\rho_a/(1 - 4\rho_a) < q < 1 - 4\rho_a/(1 - 4\rho_a)$ of Corollary 3.6, which is contained in the interval. Lower row: g_a for $a = 0.64$, $a = a_*$, and $a = 0.72$. The top at a_* is quartic by Lemma B.2.

bias we do the same with first differences (Figure 2). By Proposition 2.5, the first differences within this longer row swap signs at most 3 times. By Lemma A.1, all these bounds pass to g_a, g_a', and g_a''.

Let $q \geq 1/2$. Proposition A.3 eliminates the possibility of a $-$ and then a $+$ on the right half. So on $[0, \infty)$ the density $g_{a,q}$ has no strict valley, and $g_{a,q}(-x) \leq g_{a,q}(x)$. Hence $g_{a,q}$ has two peaks if and only if $g_{a,q}(x) > g_{a,q}(y)$ for some $x < y \leq 0$. For fixed x, y this difference is affine in q and is ≤ 0 at $q = 1$. If it is positive at some q, it is positive on $[1/2, q]$. Taking the union over x, y and reflecting gives an interval $(b_a, 1 - b_a)$ (Theorem 3.3). By Lemma A.5, $h_a'(0) > 0$ for every $a \in (1/2, 1)$, and we can conclude $b_a > 0$.

For a_*, let E be an independent exponential variable. The moments of $E^a \mathcal{L}_a$, where $\mathcal{L}_a$ is the limit of S_N/N^a given $X_1 = 1$, give the transform of [12, Proposition 3.2]. Two integrations by parts in its Mellin transform give $g_a''(0)$ as a positive multiple of $\mathcal{R}(a) - 1$, where $\mathcal{R}(a)$ is a weighted mean against a positive weight (Lemma B.1). Rational bounds give the sign of $g_a''(0)$ for $a \notin [2/3, 69/100]$. On $[2/3, 69/100]$ the score of the weight is increasing in the integration variable, and a covariance identity gives $\mathcal{R}'(a) > 0$ (Appendix B). Hence $g_a''(0)$ changes sign exactly once as a increases (Theorem 3.2). The boundary histories and this positive-weight integral are the new parts of the argument.

1.3 Prior work

The counting chain predates the elephant walk. The Drezner–Farnum model [2], in the formulation of James, James, and Qi [4, p. 2339], has next-success probability $(1-\theta)p+\theta k/t$. With $p=1/2$ and $\theta=a$ this is $r_t(k)$. Their family contains bimodal laws. Heyde [3] gives the limit regimes at fixed parameters, and James, James, and Qi allow θ to depend on t. Baur and Bertoin [6] prove functional limit theorems through Pólya-type urns. Its memory structure is a random recursive tree [7], and the split of this tree into two subtrees is used in Appendix A. Zeros and recurrence of the walk are studied in [8, 9]. See also [5, 10, 11].

We use the density and moments from [13, Theorems 1.3 and 1.4] and conditional unimodality from [12, Theorem 1.1]. Kiss and Vető [14] compute the first moments for general steps. Transition probabilities, Gaussian rates, and moment recurrences are in the preprints [15, 18, 19]. Trinas and Valle [16] prove a superdiffusive local limit theorem, and show the conditional law is not log-concave. Their difference recurrence also gives Lemma A.1. Since their paper is a preprint, we include the proof. Trinas [17] uses their local limit theorem for the last-return law, in which the value $g_{a,q}(0)$ appears.

Da Silva, Cressoni, Schütz, Viswanathan, and Trimper [23] showed that the superdiffusive propagator is not Gaussian. In a growing-network model, Mohandas et al. [24, Section V and Figure 5] found the fair-start transition numerically. They extrapolated finite-N central curvatures and got copying probability about 0.836. In their parameter, a_* corresponds to $(1+a_*)/2\approx 0.8344$.

The weights giving two peaks are known for normal mixtures [27, 28], but h_a is not Gaussian, and the interior step of our induction is classical variation diminution [25, 26].

2 Finite shape bounds

This section proves the finite bounds behind Theorems 1.1 and 3.1. For a fair start, Theorem 2.4 shows that the second difference of the count law changes sign at most four times. For every initial bias, Proposition 2.5 shows that the count law has at most two modal plateaus. The limit transfer is Lemma A.1 in Appendix A.

We first treat second differences under a fair start. Let $0<\varepsilon<\frac12$, $a=1-2\varepsilon$, and $r_t(k)=\varepsilon+ak/t$. For $t\ge 1$ and every integer k we write $P_t(k)=\mathbb{P}(A_t=k)$, which is 0 unless $0\le k\le t$, and

$$D_t(k)=P_t(k+1)-2P_t(k)+P_t(k-1).$$

We also put $s_t=D_t(1)=P_t(0)-2P_t(1)+P_t(2)$ for $t\ge 2$.

Lemma 2.1 (recurrence for the second difference)**.** *For every $t\ge 1$ and every integer k,*

$$P_{t+1}(k)=\big(1-r_t(k)\big)P_t(k)+r_t(k-1)\,P_t(k-1), \tag{2.1}$$

$$D_{t+1}(k)=\big(1-r_t(k+1)\big)D_t(k)+r_t(k-2)\,D_t(k-1). \tag{2.2}$$

Also $P_t(k)=P_t(t-k)$ and $D_t(k)=D_t(t-k)$.

Proof. Given $A_t=k$, the copied step is $+1$ with probability k/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{k}{t}+\varepsilon(1-\frac{k}{t})=r_t(k)$. This gives (2.1) for $0\le k\le t+1$. At $k=0$ and $k=t+1$, the extra term $r_t(-1)P_t(-1)$ or $(1-r_t(t+1))P_t(t+1)$ is 0 because $P_t(-1)=P_t(t+1)=0$. Outside $0\le k\le t+1$ both sides are 0. Under a fair start, (X_i) and $(-X_i)$ have the same law, so $P_t(k)=P_t(t-k)$. Taking second differences gives the same for D_t.

For (2.2), take second differences in (2.1). The affine identity $r_t(k+1)-r_t(k)=a/t$ gives

$$2r_t(k+1)+r_t(k-2)=3r_t(k),\qquad r_t(k+1)+2r_t(k-2)=3r_t(k-1),$$

so the coefficients of $P_t(k)$ and $P_t(k-1)$ agree with the right side of (2.2). The outer two coefficients agree directly. □

Trinas and Valle [16, (2.2)] use (2.1), the forward equation, for a superdiffusive local limit theorem. Their higher-difference recurrence [16, (4.10)] is the basis of a computer-certified study of log-concavity at fixed memory. We use (2.2) for a different purpose, to count sign changes. For $2 \le k \le t-1$ its coefficients are positive. So new changes can only come in at the ends. Lemma 2.2 shows that positive sums of neighbors never add a change (see [25, 26] for the general theory).

Let z be a finite real list. Delete its zeros, replace each entry left by its sign, and merge equal neighbors. What remains is the sign pattern $\mathrm{sp}(z)$, an alternating list of signs. For two such lists we write $\sigma \preceq \tau$ if σ is a subsequence of τ. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 2.2 (positive sums of neighbors)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \le i \le n-1$ with $A_i \ge 0$ and $B_i \ge 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ and $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.*

Proof. Insert each η_i between its two neighbors in ζ. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The same holds for $\eta_i < 0$, proving the first claim. For the second, obtain η from w by deleting the entries ζ_i. Deletion can only remove signs from the pattern. □

For the walk we apply Lemma 2.2 to the augmented rows of Figure 2, drawn there for first differences. Proposition 2.5 treats the whole row and Proposition A.3 the half $2k+1 \ge t$.

$$\underbrace{\ell_N, \dots, \ell_{t+1}}_{\text{left boundary history}} \qquad \underbrace{d_t(0), \dots, d_t(t-1)}_{\text{active row}} \qquad \underbrace{u_{t+1}, \dots, u_N}_{\text{right boundary history}}$$

$$\downarrow \text{ copy or sum adjacent entries}$$

$$\ell_N, \dots, \ell_{t+2} \qquad \ell_{t+1}, d_{t+1}(1), \dots, d_{t+1}(t-1), u_{t+1} \qquad u_{t+2}, \dots, u_N$$

Figure 2: One step of the augmented rows for a final length N, with $d_t(k) = P_t(k+1) - P_t(k)$, $\ell_t = d_t(0)$, and $u_t = d_t(t-1)$. The active row gains an entry and the histories on each side lose one. Each new end value was already in a history, so the bound for the previous row already counts its sign.

Lemma 2.3 (sign changes next to an end)**.** *For every $t_1 \ge 2$ we have $\mathrm{sp}(s_2, s_3, \dots, s_{t_1}) \preceq \Pi_3$.*

We prove this in Appendix C.

Theorem 2.4 (at most five blocks)**.** *Let $N \ge 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\mathrm{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \dots, N-1\}$.

Proof. For $3 \le t \le N$, we retain the endpoint history in the list $\mathcal{Z}_t$

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N-t-1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

To apply Lemma 2.2, we check how $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ for $3 \le t \le N-1$. The first $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N-t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t-2$ entries are $D_{t+1}(j)$ for $2 \le j \le t-1$. By (2.2),

$$D_{t+1}(j) = r_t(j-2)\, D_t(j-1) + \big(1 - r_t(j+1)\big) D_t(j),$$

and $D_t(j-1)$ and $D_t(j)$ are neighbors in the middle part of $\mathcal{Z}_t$, in the required order. The coefficients satisfy $r_t(j-2) \ge \varepsilon$ and $1 - r_t(j+1) \ge 1 - \varepsilon - a = \varepsilon$, since $2 \le j \le t-1$. The last $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 2.2, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 2.3 the sign pattern p of $(s_3, \dots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$, and $(+,-,+)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$, and $(+,-,+,-,+)$, and each is a subsequence of Π_5. □

2.1 First differences for a biased start

Proposition 2.5 (two modes for every initial bias)**.** *Let $0 < \varepsilon < 1$, let $q \in [0,1]$ be the probability that the first step is positive, and let $P_n(k)$ be the law of the number of positive steps after $n \ge 1$ steps. Extend P_n by zero outside $\{0, \dots, n\}$. The first differences $d_n(k) = P_n(k+1) - P_n(k)$ satisfy*

$$\mathrm{sp}\big(d_n(-1), d_n(0), \dots, d_n(n)\big) \preceq (+,-,+,-).$$

In particular, P_n has at most two modal plateaus, including endpoint modes. Moreover, there are no integers $i_0 < \cdots < i_4$ for which

$$P_n(i_0) > P_n(i_1) < P_n(i_2) > P_n(i_3) < P_n(i_4).$$

Proof. Put $a = 1 - 2\varepsilon$ and $r_t(k) = \varepsilon + ak/t$. The forward equation is

$$P_{t+1}(k) = (1 - r_t(k))P_t(k) + r_t(k-1)P_t(k-1), \qquad P_1 = (1-q, q).$$

Subtracting adjacent equations gives

$$d_{t+1}(k) = r_t(k-1)d_t(k-1) + (1 - r_t(k+1))d_t(k), \qquad 1 \le k \le t-1. \tag{2.3}$$

Both coefficients are at least $\min(\varepsilon, 1-\varepsilon) > 0$, since $r_t(j)$ is a convex combination of ε and $1-\varepsilon$ for $0 \le j \le t$. It remains to check the two ends.

Suppose first that $q < 1$. Since $P_t(0) = (1-q)(1-\varepsilon)^{t-1} > 0$, define

$$\xi_t = \frac{P_t(1)}{P_t(0)}, \qquad b = \frac{1-2\varepsilon}{1-\varepsilon} < 1, \qquad A = \frac{\varepsilon}{1-\varepsilon} > 0.$$

The boundary equation gives

$$\xi_{t+1} = (1 - b/t)\xi_t + A, \qquad \xi_1 = \frac{q}{1-q}.$$

Thus $\delta_t = \xi_{t+1} - \xi_t$ obeys

$$\delta_{t+1} = \frac{t-b}{t+1}\delta_t + \frac{A}{t+1}. \tag{2.4}$$

Once an increment is nonnegative, every later increment is positive. Thus ξ_t first decreases and then increases (either part may be empty). For $\ell_t = d_t(0) = P_t(0)(\xi_t - 1)$ this proves

$$\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (+, -, +).$$

If $q \le 1/2$, then $\xi_1 \le 1$ and the sharper bound is $\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (-, +)$. In particular, a positive ℓ_i cannot be followed by a negative ℓ_j. Zeros do not affect either conclusion.

Write $u_t = d_t(t-1)$. Reflection of the walk gives $u_t(q) = -\ell_t(1-q)$ and $d_t^{(1-q)}(t-1-k) = -d_t^{(q)}(k)$. For $0 < q \le 1/2$, the right boundary history therefore has pattern contained in $(-, +, -)$. For $q = 0$ it is nonpositive throughout. The conditional count is at most $t - 1$, so $u_t = -P_t(t-1) \le 0$.

We put the values arriving at the boundary in an augmented row, since they are not controlled by the interior recurrence. Fix N and, for $1 \le t \le N$, form

$$Z_t = (\ell_N, \dots, \ell_{t+1}, d_t(0), \dots, d_t(t-1), u_{t+1}, \dots, u_N).$$

The left history is read backwards while the right is read forwards, so every endpoint that arrives was already in the previous augmented row and the recurrence ends up controlling the entire row. For $q \le 1/2$, its first row is contained in $(+, -, +, -)$. We must also allow zeros at $\ell_1 = u_1 = 2q - 1$. Suppose Z_1 contained four ordered signs $-, +, -, +$. If the second lay in the reversed left history, the first two would give a positive left edge followed in time by a negative one, which is impossible. Otherwise the last three lie in the right history and have signs $+, -, +$. For $q > 0$, their reflected negatives give the forbidden pattern $(-, +, -)$ in the left history at bias $1 - q$. For $q = 0$, they contradict the nonpositive right history. This also covers $N = 1$.

Every entry of Z_{t+1} is a nonnegative combination of adjacent entries of Z_t. The interior entries use (2.3). The two new boundary entries and the remaining future edge values are copied, with coefficients $(1, 0)$ or $(0, 1)$. To check the ordering, number the augmented entries from left to right. Output entry i uses input entries i and $i + 1$, with a zero coefficient when copied. A nonzero output has a parent of the same sign. Two neighbors in the subsequence with opposite signs cannot share a parent, since one entry has only one sign. So the parent indices strictly increase. Pick parents like this for an alternating subsequence of the output. We get the same signs, in the same order, in the input. An update therefore cannot add an alternating pattern, so $\operatorname{sp}(Z_N) \preceq (+, -, +, -)$. Reflection and sign reversal preserve this pattern, proving the claim also for $q \ge 1/2$, including $q = 1$.

Finally, extension by zero adds $d_N(-1) = P_N(0) \ge 0$ on the left and $d_N(N) = -P_N(N) \le 0$ on the right. Neither enlarges the pattern. Every modal plateau contributes a positive-to-negative change, so there are at most two. If the statement's last inequalities held, telescoping on each of their four intervals would give an adjacent difference of the same strict sign. These four differences would have signs $-, +, -, +$, a contradiction. □

3 The limit density at fixed memory

Fix $\varepsilon \in (0, \frac{1}{4})$, so $a = 1 - 2\varepsilon \in (\frac{1}{2}, 1)$. Conditional on $X_1 = 1$, S_k/k^a converges almost surely to $\mathcal{L}_a$ [10, Theorem 3.7]. The fair-start limit $\mathcal{L}_a^\circ$ has the law of $\xi\mathcal{L}_a$, where ξ is a fair sign independent of $\mathcal{L}_a$ [13, (1.3) and (1.6)]. Guérin, Laulin, and Raschel prove that both limits have positive, bounded, and smooth densities [13, Theorem 1.3], denoted by h_a and g_a as before. Guérin, Laulin, Raschel, and Simon prove that $\mathcal{L}_a$ is unimodal [12, Theorem 1.1]. We also use two regularity facts, proved in Lemma D.1 of Appendix D. For every bias q, the limit density extends to an entire function, so it is not constant on

any interval. For every $k \geq 0$, the derivatives $\partial_u^k h_a(u)$ and $\partial_a \partial_u^k h_a(u)$ exist and are jointly continuous in (a, u).

3.1 Modes and inflection points

A probability density h on $\mathbb{R}$ is *unimodal* if for some m it is nondecreasing on $(-\infty, m]$ and nonincreasing on $[m, \infty)$. Guérin, Laulin, Raschel, and Simon ask for a unique local maximum [12], so we must also exclude a flat top. The density g_a mixes the unimodal density of $\mathcal{L}_a$ with its reflection. Such a mixture can have more than one peak [22]. We first take a fair start. Lemma D.1 excludes constant intervals, so modes are points. We consider other biases afterward.

Theorem 3.1 (at most two modes)**.** *Let $a \in (\frac{1}{2}, 1)$. Exactly one of the following holds.*

(a) The density g_a is nondecreasing on $(-\infty, 0]$ and nonincreasing on $[0, \infty)$.

(b) There is $\kappa > 0$ such that g_a is nondecreasing on $[0, \kappa]$ and nonincreasing on $[\kappa, \infty)$, and $g_a(0) < g_a(\kappa)$.

In case (b), g_a is nondecreasing on $(-\infty, -\kappa]$ and nonincreasing on $[-\kappa, 0]$, and it is not unimodal. All these monotonicities are strict. The unique local maximum is 0 in case (a), and the only local maxima are $\pm\kappa$ in case (b). In case (a), there is $d > 0$ such that g_a is strictly concave on $(-d, d)$ and strictly convex on each tail. In case (b), there are $0 < d_1 < \kappa < d_2$ such that g_a is strictly convex on $(-d_1, d_1)$ and on each tail beyond $\pm d_2$, and strictly concave on $(-d_2, -d_1)$ and (d_1, d_2). Thus it has exactly two inflection points in case (a) and four in case (b).

A real function v has *values at most* Π if $\operatorname{sp}(v(y_1), \dots, v(y_j)) \preceq \Pi$ for every $y_1 < \cdots < y_j$.

Proof. Fix $a \in (1/2, 1)$. Lemma A.1 at order two passes the five-block bound of Theorem 2.4 to g_a''. Indeed, the interpolant's values are positive sums of adjacent second differences, and its lattice fills every fixed compact interval as $N \to \infty$. The four values $D_N(0)$, $D_N(-1)$, $D_N(N)$, and $D_N(N+1)$ near the ends sit at the points $(2j - N)N^{-a}$ with $j \in \{-1, 0, N, N+1\}$, since $\nabla^2 P_N(k) = D_N(k-1)$. Their absolute values are at least N^{1-a}. Since $a < 1$, these points leave every compact interval. Hence on a fixed compact interval the interpolant uses only interior second differences for large N. Lemmas 2.2 and A.2 therefore give values at most Π_5.

Symmetry. Proposition A.3 and Lemma A.1, with $r = 0$, show that g_a has no strict valley on $[0, \infty)$. Such a valley would persist at nearby lattice points under locally uniform convergence.

If every increment on $[0, \infty)$ is nonpositive, case (a) holds. Otherwise choose $y_1 < y_2$ with $g_a(y_1) < g_a(y_2)$. Integrability supplies $y_3 < y_4$, both beyond y_2, with $g_a(y_3) > g_a(y_4)$. The no-valley property excludes any later increase, so g_a attains a maximum on $[0, \infty)$ at some κ. The absence of a strict valley makes it nondecreasing before κ and nonincreasing afterward. The positive increment forces $\kappa > 0$. If $g_a(0) = g_a(\kappa)$, it would be constant on $[0, \kappa]$, contrary to analyticity and the positive increment. Thus case (b) holds. Evenness gives the left-half shape and excludes unimodality in (b).

Equality at two points of a monotone interval would force a constant interval, contrary to Lemma D.1. Thus the monotonicities are strict and there is no additional local maximum.

Curvature. The Fourier bound in Lemma D.1 gives $g_a(y), g_a'(y) \to 0$ as $y \to \infty$. Evenness and the Π_5 bound give at most three sign blocks for g_a'' on $(0, \infty)$. Its last sign must be positive. Otherwise g_a' would eventually be nonincreasing with limit zero, making g_a eventually nondecreasing. This contradicts $g_a > 0$ and $g_a \to 0$. Analyticity and

strict monotonicity imply that the first nonconstant term in the even Taylor series of g_a at zero is negative in case (a) and positive in case (b). The same signs therefore hold for g_a'' near zero. In case (b), g_a'' must also be negative somewhere, since g_a' is positive near zero and negative somewhere beyond κ. The positive-half-line sign patterns are consequently $(-,+)$ in case (a) and $(+,-,+)$ in case (b). Analyticity excludes an interval of zero curvature and makes the resulting convexities strict. In case (b), $g_a'(d_1) > 0 > g_a'(d_2)$, so $d_1 < \kappa < d_2$. □

Theorem 3.2 (the unique transition)**.** *There is exactly one $a_* \in (1/2, 1)$ with $g_{a_*}''(0) = 0$. It lies in $(2/3, 69/100)$, and*

$$g_a''(0) < 0 \quad (1/2 < a < a_*), \qquad g_a''(0) > 0 \quad (a_* < a < 1).$$

Moreover, $\partial_a g_a''(0)|_{a=a_*} > 0$. The density g_a is unimodal for $a \le a_*$ and bimodal for $a > a_*$. In particular, the critical density is unimodal.

To rule out a second crossing, Appendix B expresses the curvature sign as a comparison of positive weighted means, without numerical input. Rational bounds fix the sign outside $[2/3, 69/100]$. On this interval the score is increasing, and a covariance identity shows that the weighted mean is strictly increasing in a. Hence there is exactly one crossing. Numerically $a_* \approx 0.66881$ (Remark 1.4).

3.2 Arbitrary initial bias

Write h_a for the density of $\mathcal{L}_a$ and $g_{a,q}(x) = qh_a(x) + (1-q)h_a(-x)$ for initial bias $\mathbb{P}(X_1 = 1) = q$. Thus $g_{a,1/2} = g_a$.

Theorem 3.3 (the interval of bimodal biases)**.** *For every $a \in (1/2, 1)$ and $q \in [0, 1]$, the density $g_{a,q}$ has at most two local maxima. There is $b_a \in (0, 1/2]$ such that it has exactly two local maxima if and only if*

$$b_a < q < 1 - b_a.$$

Outside this open interval, including its endpoints, the density is unimodal with a unique local maximum. The interval is empty exactly when $a \le a_*$, in which case $b_a = 1/2$. Positivity of b_a for $a > a_*$ is proved in Theorem 3.5. At finite time, for every $0 < \varepsilon < 1$, $q \in [0, 1]$, and $N \ge 1$, the count law has at most two modal plateaus, including its endpoints.

Proof. The mode bound. Proposition 2.5 gives the finite assertion and excludes two successive strict valleys. The locally uniform density convergence in Lemma A.1, with $r = 0$, passes these inequalities to $g_{a,q}$. Lemma D.1 excludes constant intervals for every bias. Hence three local maxima $x < y < z$ would give $x < u < y < v < z$ with

$$g_{a,q}(u) < \min\{g_{a,q}(x), g_{a,q}(y)\}, \qquad g_{a,q}(v) < \min\{g_{a,q}(y), g_{a,q}(z)\}.$$

For each pair, take a one-sided neighborhood of the smaller maximum toward the other one. Since the density is not constant on an interval, some point there has strictly smaller value. This gives the two forbidden valleys.

The bias interval. Fix a, write $f_q = g_{a,q}$ and $h = h_a$, and call $x < y < z$ a strict valley if $f_q(y) < \min\{f_q(x), f_q(z)\}$. Proposition A.3 and the same density convergence show that, for $q \ge 1/2$, f_q has no strict valley on $[0, \infty)$ and $f_q(-x) \le f_q(x)$ for $x \ge 0$. Thus it has a strict valley if and only if $f_q(x) > f_q(y)$ for some $x < y \le 0$. Reflection dominance turns such a descent into the valley $(x, y, -x)$. Conversely, a valley must start at $x < 0$. If its middle point y is positive, the no-valley property at $(0, y, z)$ gives $f_q(0) \le f_q(y) < f_q(x)$.

At $q = 1$, conditional unimodality [12] and reflection dominance place the mode in $[0, \infty)$. A negative mode would have a reflected point of at least the same height,

contrary to uniqueness. Thus h is nondecreasing on $(-\infty, 0]$. For fixed $x < y \le 0$, the difference $f_q(x) - f_q(y)$ is affine in q and nonpositive at $q = 1$. If positive at $q \ge 1/2$, it remains positive for every $q' \in [1/2, q]$. Thus, within $[1/2, 1]$, a valley persists when the bias decreases. These biases form a relatively open set because the strict inequalities at three fixed points survive a small change of bias. The set excludes 1, and reflection gives an open interval $(b_a, 1 - b_a)$, possibly empty.

A strict valley gives two local maxima. Integrability and positivity let us choose an outer point on each side below the corresponding high point. We then maximize on the two resulting compact intervals. Conversely, two local maxima give a strict valley because there is no flat interval. The mode bound excludes a third maximum. A continuous integrable density without a strict valley is unimodal, and absence of flat intervals makes its maximum unique. Finally the bias interval is nonempty exactly when it contains $1/2$, so Theorem 3.2 identifies the empty case. □

This proves the initial-bias conjecture of [12, Remark 2.1(b)], with its half-width $q(a) = 1/2 - b_a$. The derivative-limit argument used here is proved in Appendix A.

3.3 The bias boundary

For $a > a_*$, Theorem 3.2 puts g_a in case (b) of Theorem 3.1. We write κ_a for its positive mode κ.

To find the endpoint, we compare the conditional density's slopes at reflected points. Their ratio tells us how much weight the reflected density needs to create a decrease.

Proposition 3.4 (a variational equation for the bias boundary)**.** *For $a \in (1/2, 1)$ put $U_a = \{t > 0 : h_a'(t) > 0\}$. With the infimum over an empty set interpreted as $+\infty$,*

$$b_a = \min\left\{\frac{1}{2}, \quad \inf_{t \in U_a} \frac{h_a'(-t)}{h_a'(t) + h_a'(-t)}\right\}. \tag{3.1}$$

If the infimum is attained at $t > 0$ with $h_a'(t) > 0$ and $b_a < 1/2$, then, with $q = 1 - b_a$,

$$g_{a,q}'(-t) = g_{a,q}''(-t) = 0.$$

At this point the smaller peak disappears.

Proof. The proof of Theorem 3.3 shows that $h_a'(-t) \ge 0$ for $t \ge 0$. For $q \ge 1/2$ bimodality is equivalent to a decrease somewhere on the negative half-line, hence to $g_{a,q}'(-t) < 0$ for some $t > 0$. Here

$$g_{a,q}'(-t) = q h_a'(-t) - (1 - q) h_a'(t).$$

This expression can be negative only if $t \in U_a$, in which case it is negative exactly when $1 - q > h_a'(-t)/(h_a'(t) + h_a'(-t))$. Taking the union over t gives (3.1), with equality excluded as in Theorem 3.3. At an interior minimizer the ratio has derivative zero. Differentiating its numerator and denominator gives the two displayed equations. □

Theorem 3.5 (regularity of the bias boundary)**.** *The function $a \mapsto b_a$ is continuous on $(1/2, 1)$, is continuously differentiable on $(a_*, 1)$, and extends continuously to $b_1 = 0$. It is strictly decreasing on some interval immediately above a_*. For every $a > a_*$, we have $0 < b_a < 1/2$, and the infimum in (3.1) has a unique minimizer $t_a \in (0, \kappa_a)$. This minimizer depends continuously on a throughout $(a_*, 1)$.*

Proof. For $a > a_*$, the curvature partition in Theorem 3.1 gives $g_a'(t) > 0$ on $(0, \kappa_a)$ and $g_a'(t) < 0$ on (κ_a, ∞). For $t \ge \kappa_a$ in U_a, the ratio in (3.1) is therefore at least $1/2$. Since

$b_a < 1/2$, it is enough to search $(0, \kappa_a)$ (we still have to show the infimum is attained). Put

$$D_a(t) = h'_a(t) + h'_a(-t), \qquad R_a(t) = \frac{g'_a(t)}{D_a(t)} \quad (0 < t < \kappa_a).$$

Lemmas A.4 and A.5 give $h'_a(-t) > 0$ for every $t \geq 0$. We therefore have $D_a > 0$, $0 < R_a < 1/2$, and $1/2 - b_a = \sup_{0<t<\kappa_a} R_a(t)$. Also $R_a(0+) = 0$.

We first rule out a strict valley in the ratio. We claim that

$$R_a(v) \geq \min\{R_a(u), R_a(w)\} \qquad (0 < u < v < w < \kappa_a). \tag{3.2}$$

Otherwise, choose $s \in (0, 1/2)$ strictly between the middle value and both outer values. With $q = 1/2 + s$, $g'_{a,q}(-t) = D_a(t)(s - R_a(t))$ then has signs $-, +, -$ at $-w, -v, -u$, and is positive sufficiently close to zero on the left. Positivity of the density and its zero limits at infinity supply a positive derivative farther left and a negative derivative farther right. There would be three local maxima, contrary to Theorem 3.3.

At κ_a, we have $D_a(\kappa_a) = 2h'_a(-\kappa_a) > 0$. Thus R_a extends continuously to $[0, \kappa_a]$, with value zero at both endpoints. Its positive maximum is attained in the interior. At that point $h'_a(-t) > 0$, so the maximum is strictly below $1/2$. Hence $b_a > 0$. We need attainment here (pointwise strict inequality on an open interval would not suffice). Equation (3.2) and analyticity make this maximum unique, since two maximizers would force a constant interval. They also make R_a strictly increasing before its maximizer and strictly decreasing after it.

Lemma D.1 gives joint continuity of all the derivatives used here. The signs of g'_a on either side of its unique positive zero give continuity of κ_a. Fix $a_0 > a_*$ and choose $0 < l < t_{a_0} < r < \kappa_{a_0}$. The strict inequalities $R_a(l), R_a(r) < R_a(t_{a_0})$ persist near a_0. For $v < l$, apply (3.2) to v, l, t_{a_0}. Since $R_a(l) < R_a(t_{a_0})$, it gives $R_a(v) \leq R_a(l)$. Applying it to t_{a_0}, r, v gives $R_a(v) \leq R_a(r)$ for $v > r$. Thus every nearby global maximum lies in the fixed interval $[l, r]$. The ratios converge uniformly on this fixed compact interval, so their maxima converge. Uniqueness gives convergence of the maximizers too. This proves continuity above a_*.

At a_*, the curvature partition in Theorem 3.1(a) makes g'_{a_*} strictly decrease from zero and then strictly increase to zero. Thus $g'_{a_*}(r) < 0$ for every $r > 0$. Joint continuity gives $g'_a(r) < 0$ for all $a > a_*$ sufficiently close to a_*, so $\kappa_a < r$. Since r was arbitrary, $\kappa_a \to 0$. On the shrinking interval $[0, \kappa_a]$, joint continuity now gives

$$\sup_{0 \leq t \leq \kappa_a} |g'_a(t)| \longrightarrow 0, \qquad \inf_{0 \leq t \leq \kappa_a} D_a(t) \longrightarrow 2h'_{a_*}(0) > 0,$$

where the strict inequality is Lemma A.5. Consequently $1/2 - b_a = \max R_a \to 0$. Below the threshold $b_a = 1/2$, so this proves continuity at a_* without the critical expansion. Corollary 3.6 gives the limit at one.

For differentiability, the same three-point argument puts all nearby maximizers in one compact interval. If $M(a) = 1/2 - b_a$, maximality gives, for $s > 0$,

$$R_{a+s}(t_a) - R_a(t_a) \leq M(a+s) - M(a) \leq R_{a+s}(t_{a+s}) - R_a(t_{a+s}).$$

Divide by s and use the continuity of t_a and $\partial_a R_a$ from Lemma D.1. The argument for $s < 0$ is the same. Thus

$$-b'_a = \frac{\partial_a h'_a(t_a)\, h'_a(-t_a) - h'_a(t_a)\, \partial_a h'_a(-t_a)}{\{h'_a(t_a) + h'_a(-t_a)\}^2}. \tag{3.3}$$

The right side is continuous. No derivative of t_a is needed.

Finally, put $C_* = \partial_a g_a''(0)|_{a=a_*} > 0$ and $H_* = h'_{a_*}(0) > 0$, by Theorem 3.2 and Lemma A.5. Extend $R_a = g_a'/D_a$ to a fixed neighborhood of zero, where $D_a > 0$ for a close to a_*. Evenness and Lemma D.1 give, jointly as $(a,t) \to (a_*, 0)$,

$$g_a'(t) = o(t), \qquad \partial_a g_a'(t) = (C_* + o(1))t, \qquad D_a(t) = 2H_* + o(1), \qquad \partial_a D_a(t) = O(1).$$

For the first two relations, integrate g_a'' and $\partial_a g_a''$ from zero to t, using $g''_{a_*}(0) = 0$. It follows that

$$\partial_a R_a(t) = \left(\frac{C_*}{2H_*} + o(1)\right) t > 0$$

for all sufficiently small $t > 0$ and a sufficiently close to a_*. Since $0 < t_a < \kappa_a \to 0$, the envelope formula gives $b_a' < 0$ on some interval immediately above the threshold. □

Corollary 3.6 (a rate for the bias boundary near one)**.** *Put*

$$\rho_a = \frac{\Gamma(1+a)^2}{(2a-1)\Gamma(2a)} - 1.$$

If $\rho_a < 1/12$*, then*

$$b_a \le \frac{4\rho_a}{1-4\rho_a}.$$

In particular, as $a \uparrow 1$*,* $b_a \le 8(1-a) + O((1-a)^2)$*.*

Proof. The conditional limit has mean $\mu_a = 1/\Gamma(1+a) > 0$ and variance $\rho_a \mu_a^2$, by the moment formulas of [13, Theorem 1.4]. Consider the three adjacent intervals of length μ_a centered at $-\mu_a$, 0, and μ_a. Chebyshev's inequality puts at least $1-4\rho_a$ of the conditional law in the right interval and at most $4\rho_a$ in the middle. For a start to the left, reflect these intervals. After mixing, the left interval has mass at least $(1-q)(1-4\rho_a)$ and the right at least $q(1-4\rho_a)$. The middle still has mass at most $4\rho_a$.

For $4\rho_a/(1-4\rho_a) < q < 1 - 4\rho_a/(1-4\rho_a)$, the middle has less mass than either outer interval. All three intervals have the same length. So, by continuity, we can choose a point in each with the middle value strictly smaller than the other two. This rules out unimodality. The two-mode bound in Theorem 3.3 gives exactly two modes, and therefore the stated bound on b_a. Finally, the logarithmic derivative at $a = 1$ of $1 + \rho_a$ is $2\psi(2) - 2 - 2\psi(2) = -2$. Thus $\rho_a = 2(1-a) + O((1-a)^2)$, which gives the linear bound. □

The linear bound is not sharp. Conditioning on a long run of positive first steps puts most of the conditional law in a window of width of order $1-a$, and this gives a quadratic rate.

Proposition 3.7 (a quadratic rate near one)**.** *As* $a \uparrow 1$*,*

$$\limsup_{a \uparrow 1} \frac{b_a}{(1-a)^2} \le 82.$$

Proof. Let $p = 1 - \varepsilon$, fix an integer $k \ge 1$, and let $E_k = \{X_1 = \cdots = X_k = 1\}$. Given $X_1 = 1$, each of the next $k-1$ steps copies a positive step with probability $1-\varepsilon$ or flips a positive step with probability ε, so $\mathbb{P}_+(E_k) = p^{k-1}$. After time k, the step $n+1$ is $+1$ with probability $\frac{1}{2} + \frac{aS_n}{2n}$. Hence

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = \left(1 + \frac{a}{n}\right) S_n, \qquad \mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = \left(1 + \frac{2a}{n}\right) S_n^2 + 1.$$

Put $\gamma_n = \prod_{j=k}^{n-1}(1 + a/j)$ and $\beta_n = \prod_{j=k}^{n-1}(1 + 2a/j)$. Given E_k, the first relation makes S_n/γ_n a martingale with mean k, and the second gives $\mathbb{E}[S_n^2 \mid E_k] = \beta_n\big(k^2 + \sum_{m=k+1}^{n} \beta_m^{-1}\big)$. The identity

$$(2a-1)\frac{\Gamma(m)}{\Gamma(m+2a)} = \frac{\Gamma(m)}{\Gamma(m+2a-1)} - \frac{\Gamma(m+1)}{\Gamma(m+2a)}$$

telescopes to $\sum_{m>k}\beta_m^{-1} = k/(2a-1)$. Since $\gamma_n/n^a \to \Gamma(k)/\Gamma(k+a)$ and $\beta_n/\gamma_n^2 \to c_k = \Gamma(k+a)^2/(\Gamma(k)\Gamma(k+2a))$, the martingale is bounded in L^2. Its limit gives, given E_k,

$$\mathbb{E}[\mathcal{L}_a \mid E_k] = m_k = \frac{\Gamma(k+1)}{\Gamma(k+a)}, \qquad \mathrm{Var}(\mathcal{L}_a \mid E_k) = r_k m_k^2, \qquad r_k = c_k\Big(1 + \frac{1}{(2a-1)k}\Big) - 1.$$

For $k=1$ this is $r_1 = \rho_a$. To bound r_k, let $\varphi(s)$ be $\log(1+r_k)$ with a replaced by s. Then $\varphi(1) = 0$ and

$$\varphi'(s) = -2\big(\psi(k+2s) - \psi(k+s)\big) - \frac{2}{(2s-1)\{(2s-1)k+1\}} < 0.$$

Since $\psi'(x) < 1/(x-1)$ for $x > 1$, we have $\psi(k+2s) - \psi(k+s) \le s\psi'(k+s) \le 1/k$ for $s \le 1$. Also $(2s-1)k + 1 \ge (2s-1)(k+1)$. So $0 \le \log(1+r_k) \le (1-a)\theta_k$ with $\theta_k = 2/k + 2/((2a-1)^2(k+1))$.

Now fix $K > 1$, put $v = \sqrt{r_k}$, and assume $3Kv < 1$ and $\pi = p^{k-1}(1-K^{-2}) > 1/2$. Let $x_0 = m_k(1-Kv)$ and $\ell = 2Km_k v$, so $x_0 - \ell = m_k(1-3Kv) > 0$. By Chebyshev's inequality given E_k, the window $W = [x_0, x_0+\ell]$ has $\mathbb{P}_+(\mathcal{L}_a \in W) \ge \pi$, and so $\mathbb{P}_+(\mathcal{L}_a < x_0) \le 1-\pi$. By Lemmas A.4 and A.5, h_a is increasing on $(-\infty, 0]$. So for $y < 0$ we have $|y|h_a(y) \le \mathbb{P}_+(\mathcal{L}_a \le 0)$, and Cantelli's inequality gives $\mathbb{P}_+(\mathcal{L}_a \le 0) \le \rho_a/(1+\rho_a)$. On $J_2 = [-x_0, -x_0+\ell]$ this gives

$$\mathbb{P}_+(\mathcal{L}_a \in J_2) \le \frac{\ell}{x_0 - \ell} \cdot \frac{\rho_a}{1+\rho_a} = \frac{2Kv}{1-3Kv} \cdot \frac{\rho_a}{1+\rho_a} =: A.$$

Let $q \in [1/2, 1)$ and $J_1 = [-x_0 - \ell, -x_0]$, so $-J_1 = W$ and $-J_2 \subset (-\infty, x_0]$. Then

$$\begin{aligned}\int_{J_1} g_{a,q} - \int_{J_2} g_{a,q} &\ge -q\,\mathbb{P}_+(\mathcal{L}_a \in J_2) + (1-q)\big(\mathbb{P}_+(\mathcal{L}_a \in W) - \mathbb{P}_+(\mathcal{L}_a < x_0)\big) \\ &\ge -qA + (1-q)(2\pi - 1).\end{aligned}$$

If $1 - q > A/(2\pi - 1 + A)$, the right side is positive. The intervals J_1 and J_2 are adjacent, have equal length, and lie in $(-\infty, 0)$ with J_1 on the left, so $g_{a,q}$ is not nondecreasing on $(-\infty, 0]$. Thus $g'_{a,q}(-t) = qh'_a(-t) - (1-q)h'_a(t) < 0$ for some $t > 0$. Since $h'_a(-t) > 0$, this forces $t \in U_a$ and $h'_a(-t)/(h'_a(t) + h'_a(-t)) < 1 - q$. By (3.1), $b_a < 1 - q$, and therefore

$$b_a \le \frac{A}{2\pi - 1 + A} \le \frac{A}{2\pi - 1}.$$

Finally let $a \uparrow 1$. Take $\delta = 4/25$, $K = 27/10$, and $k = \lceil 2\delta/(1-a) \rceil$. Then $p^{k-1} \to e^{-\delta}$ and $k\theta_k \to 4$, so $v^2 \le e^{(1-a)\theta_k} - 1 = (2/\delta + o(1))(1-a)^2$. With $\rho_a = 2(1-a) + O((1-a)^2)$ this gives $A = (4K\sqrt{2/\delta} + o(1))(1-a)^2$ and $2\pi - 1 \to 2e^{-\delta}(1-K^{-2}) - 1 > 0$. Hence

$$\limsup_{a \uparrow 1} \frac{b_a}{(1-a)^2} \le \frac{4K\sqrt{2/\delta}}{2e^{-\delta}(1-K^{-2}) - 1} < 82.$$

□

3.4 The density at the threshold

How far apart are the two modes just after the transition? We compute this from derivatives at the threshold. As in the proof of Theorem 3.5, write

$$C_* = \partial_a g''_a(0)|_{a=a_*}, \qquad B_* = -g^{(4)}_{a_*}(0), \qquad H_* = h'_{a_*}(0).$$

These constants are positive by Theorem 3.2 and Lemmas B.2 and A.5. The coefficient Q_* is defined in Proposition 3.9.

Proposition 3.8 (the critical shape)**.** *The density at the threshold satisfies*

$$g_{a_*}(u) = g_{a_*}(0) - \frac{B_*}{24}u^4 + O(u^6) \qquad (u \to 0).$$

Put $K_* = \sqrt{6C_*/B_*}$*. As* $a \downarrow a_*$*, the positive mode* κ_a *of* g_a *satisfies*

$$\kappa_a = K_*\sqrt{a - a_*} + o\big(\sqrt{a - a_*}\big).$$

Proof. Lemma D.1 gives joint continuity of all spatial derivatives and their first memory derivatives. Taylor remainders through order four are therefore uniform on a neighborhood of a_*. At criticality, evenness, $g''_{a_*}(0) = 0$, and the analytic sign $g^{(4)}_{a_*}(0) = -B_* < 0$ from Lemma B.2 give the first assertion.

Put $\delta = a - a_* > 0$. Uniform Taylor expansion gives

$$\frac{g'_a(u)}{u} = C_*\delta + o(\delta) - \frac{B_* + o(1)}{6}u^2 + O(u^4).$$

For $u = k\sqrt{\delta}$ this has the sign of $C_* - B_*k^2/6$ when k is fixed away from $\sqrt{6C_*/B_*}$. The unique positive mode lies between any two such choices with opposite signs, by Theorem 3.1. Squeezing gives the mode expansion. □

Proposition 3.9 (how the bias interval opens)**.** *With* C_*, B_*, H_* *as defined above, put*

$$Q_* = \frac{\sqrt{2}\,C_*^{3/2}}{3H_*\sqrt{B_*}}.$$

As $a \downarrow a_*$*,*

$$\frac{1}{2} - b_a = Q_*(a - a_*)^{3/2} + o\big((a - a_*)^{3/2}\big). \tag{3.4}$$

The unique minimizer t_a *of Theorem 3.5 satisfies*

$$t_a = \sqrt{\frac{2C_*}{B_*}}\sqrt{a - a_*} + o\big(\sqrt{a - a_*}\big) = \frac{\kappa_a}{\sqrt{3}} + o\big(\sqrt{a - a_*}\big).$$

At $q = 1 - b_a$*, the stationary point is* $-t_a$*.*

Proof. Lemma D.1 supplies the required joint regularity. Theorem 3.2 and Lemma A.5 give $C_* > 0$ and $H_* > 0$.

Write $\delta = a - a_*$. By (3.1),

$$\frac{1}{2} - b_a = \max\left\{0, \sup_{t \in U_a} \frac{g'_a(t)}{h'_a(t) + h'_a(-t)}\right\}.$$

Theorem 3.1 makes the numerator nonpositive for $t \geq \kappa_a$. Proposition 3.8 gives $\kappa_a = O(\sqrt{\delta})$, and $H_* > 0$ makes both conditional derivatives positive on this shrinking interval. Uniform Taylor expansion gives, for bounded $v \geq 0$,

$$\frac{g'_a(v\sqrt{\delta})}{h'_a(v\sqrt{\delta}) + h'_a(-v\sqrt{\delta})} = \frac{\delta^{3/2}}{2H_*}\left(C_*v - \frac{B_*v^3}{6}\right) + o(\delta^{3/2}).$$

The cubic is largest at the unique positive point $v = \sqrt{2C_*/B_*}$. We maximize over $[0, \kappa_a/\sqrt{\delta}]$. Its right endpoint tends to $\sqrt{6C_*/B_*}$, which is past that point, and (3.4) follows. The ratio is zero at both ends, so its positive maximum is attained in $(0, \kappa_a)$. Now the convergence is uniform and the cubic has only one maximum. Hence every scaled maximizer tends to $\sqrt{2C_*/B_*}$, which is the claim for t_a. □

Both exponents can be read off from $c(t) = C_*(a - a_*)t - B_*t^3/6$, the leading part of $g'_a(t)$ near zero. Its positive zero is $\sqrt{6C_*(a - a_*)/B_*}$, the leading term of κ_a. Its maximum over $t > 0$ is $(2\sqrt{2}/3)C_*^{3/2}B_*^{-1/2}(a - a_*)^{3/2}$, and dividing by $2H_*$ gives $Q_*(a - a_*)^{3/2}$.

4 Computer assistance

Every theorem, proposition, and lemma in this paper has a written proof that uses no computer. A computer gives only the values in Remark 1.4. The repository holds interval-arithmetic certificates for a_* and for C_*, B_*, H_*, K_*, and Q_*, computed with Arb [21], together with their records, inputs, and acceptance tests. The repository also holds Lean files. They verify all finite sign arguments and the analytic threshold proof, assuming the classical weak limit as a named hypothesis, and their coverage map lists the formal statement for each result. None of the paper's proofs relies on the certificates or on the Lean files.

5 Open problems

The full bias boundary. Is b_a strictly decreasing on $(a_*, 1)$? By Theorem 3.5 it is decreasing near a_*, and Figure 1 suggests it is decreasing on the whole interval. By (3.3), $b_a' < 0$ if and only if the numerator there is positive at t_a. We cannot show this for all a.

The rate near one. Proposition 3.7 gives $b_a = O((1-a)^2)$. Is there a matching lower bound $b_a \geq c(1-a)^2$ with $c > 0$?

A finite-length threshold. For even N, does $D_N(N/2)$, as a function of a, have exactly one zero in $(0, 1)$? If so, does this zero converge to a_*? Theorem 3.2 is about the limit and does not answer this.

The Drezner–Farnum model and higher dimensions. For $p \neq 1/2$ the Drezner–Farnum rule is not $r_t(k)$, and the limit is not a mixture of one law and its reflection [2]. Hence our argument does not apply. In higher dimensions the fixed-point theory gives densities [13], but the sign counting needs a linear order.

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Code and data

The repository https://github.com/apemm/elephant-bias-interval/tree/v1.1 has the sources of this paper, the scripts that compute the numerical values in Remark 1.4 with their interval certificates and records, and the Lean files. It is archived at Zenodo, doi:10.5281/zenodo.24095945. The file certificates/data/critical_constants_certificate.json holds the enclosures of the constants.

A Derivative limits and the bias interval

A.1 Derivative limits

For $r = 0, 1$ the bounds and the local limit below are in Trinas and Valle [16, Proposition 2.1 and Theorem 3.1]. We need every r, and a fixed initial history, so we give the proof.

Lemma A.1 (derivative limits)**.** *Fix $a \in (1/2, 1)$ and a finite initial history. Extend P_n by zero outside $\{0, \dots, n\}$. Suppose the rescaled count law converges weakly to a law with smooth density f. For each fixed integer $r \geq 0$, linearly interpolate*

$$\frac{n^{(r+1)a}}{2^{r+1}} \nabla^r P_n(k) \quad \text{at} \quad (2k - n - r)n^{-a}, \qquad \nabla P_n(k) = P_n(k) - P_n(k-1).$$

These interpolants converge locally uniformly to $f^{(r)}$.

Proof. Apply ∇^r to (2.1). This gives

$$\nabla^r P_{n+1}(k) = \left(\frac{1+a}{2} - \frac{ak}{n}\right)\nabla^r P_n(k) + \left(\frac{1-a}{2} + \frac{a(k-r-1)}{n}\right)\nabla^r P_n(k-1).$$

For large n and $1 \le k \le n+r$, both coefficients are nonnegative with sum $1-(r+1)a/n$. For $k=0$ and $k=n+r+1$ one term is zero and the other has coefficient $(1+a)/2 < 1-(r+1)a/n$. Hence $\sup_k |\nabla^r P_{n+1}(k)| \le (1-(r+1)a/n)\sup_k |\nabla^r P_n(k)|$, and iterating,

$$\sup_k |\nabla^r P_n(k)| \le C_{a,r} n^{-(r+1)a}.$$

Using this at orders r and $r+1$, the interpolants are uniformly bounded and uniformly Lipschitz. So every subsequence has a further subsequence that converges locally uniformly. To find the limit, sum by parts against a smooth function with compact support and use weak convergence. What comes out is the rth distributional derivative of the limit law. Note that the lattice spacing is $2n^{-a}$, and the normalization above is exactly its $-(r+1)$ power. Since f is smooth, the distributional derivative is $f^{(r)}$. All subsequences have this same limit, and the lemma follows. □

A.2 Sign patterns

This part has the sign lemma behind Theorem 3.1. Notation is as in Sections 2 and 3.1. Deleting entries can only remove signs from a list's pattern. An alternating list p satisfies $p \preceq q$ for every alternating list q with the same first sign and at least as many entries.

Lemma A.2 (pointwise limits)**.** *Let v_n and v be real functions on $\mathbb{R}$ with $v_n(y) \to v(y)$ for every y, and let Π be an alternating list. If every v_n has values at most Π, then so has v.*

Proof. Let $y_1 < \cdots < y_j$, and let I be the set of indices i with $v(y_i) \neq 0$. For all large n, $v_n(y_i)$ has the sign of $v(y_i)$ for every $i \in I$. Then $\operatorname{sp}(v(y_1), \ldots, v(y_j))$ is the pattern of the list $(v_n(y_i))_{i\in I}$, which is a subsequence of $\operatorname{sp}(v_n(y_1), \ldots, v_n(y_j)) \preceq \Pi$. □

Proposition A.3 (the dominant half)**.** *Under the hypotheses of Proposition 2.5, suppose $q \ge 1/2$. Then $P_n(n-k) \le P_n(k)$ for $2k \ge n$, and the right half has no strict valley. That is,*

$$P_n(j) \ge \min\{P_n(i), P_n(k)\} \qquad \text{if } i < j < k \text{ and } 2i+1 \ge n.$$

Equivalently, its first differences contain no negative value followed by a positive one. Reflection gives the corresponding left-half result when $q \le 1/2$.

Proof. Write the bias explicitly as $P_{n,q}$ and put

$$\Delta_n(k) = P_{n,q}(k) - P_{n,1-q}(k) = P_{n,q}(k) - P_{n,q}(n-k).$$

The forward equation gives $\Delta_{n+1}(k) = (1-r_n(k))\Delta_n(k) + r_n(k-1)\Delta_n(k-1)$. At $n=1$, $\Delta_1(1) = 2q-1 \ge 0$. Suppose the assertion holds at length n. At a point strictly above the new center, both parents lie in the preceding upper half and both coefficients are nonnegative. At an integer center $\Delta_{n+1}(k) = 0$ by reflection. Terms outside the support vanish. This proves the induction. In particular, the central first difference of an odd row is nonnegative.

We now keep only the half favored by the start, namely the spatial indices $2k+1 \ge t$ in Z_t. At $t=1$ its pattern is contained in $(+,-)$. The central entry is $2q-1 \ge 0$, followed by the future right boundary values $u_s(q) = -\ell_s(1-q)$. Since $1-q \le 1/2$, their pattern is contained in $(+,-)$ by (2.4). We now rule out a negative difference followed by a positive

one, also among the future boundary entries. Suppose such a pair first appeared in the next row. Choose a same-sign parent for each entry. The preceding proof shows that the first parent comes before the second. The sole possible parent outside the previous half occurs at the new central difference of an odd row. That entry is nonnegative, so it cannot begin the forbidden pair. More explicitly, if a negative entry at index i of row $t+1$ satisfies $2i+1 \geq t+1$, then $2i > t$, so both its parents satisfy $2k+1 \geq t$. A positive entry further right has its parents strictly after these, as before. By induction, no restricted augmented row contains the pair. A strict valley would contain such two adjacent differences by telescoping, which proves the claim. □

A.3 A start with one step in each direction

Lemma A.4 (a balanced start)**.** *Fix the first two steps to be $+1, -1$. For $0 \leq a \leq 1$, the count law at each length $n \geq 2$ is log-concave, with positive support $\{1, \dots, n-1\}$. If $1/2 < a < 1$, its limiting density f_a^{+-} is symmetric and log-concave, with $(f_a^{+-})'(t) < 0$ for $t > 0$. Consequently the density conditional on a positive first step satisfies*

$$h_a'(t) > 0 \qquad (t < 0).$$

Proof. Starting with one step in each direction removes the boundary obstruction to the log-concavity induction of [12, Proposition 2.2]. Write one update as $g_k = Af_k + Bf_{k-1}$, where

$$A = \frac{1+a}{2} - \frac{ak}{n}, \qquad B = \frac{1-a}{2} + \frac{a(k-1)}{n}, \qquad d = \frac{a}{n}.$$

For $u = f_{k-2}$, $v = f_{k-1}$, $w = f_k$, and $z = f_{k+1}$, expansion gives

$$\begin{aligned} g_k^2 - g_{k-1}g_{k+1} = (A^2 - d^2)(w^2 - vz) + (B^2 - d^2)(v^2 - uw) \\ + (A-d)(B-d)(wv - uz) + d^2(w-v)^2. \end{aligned}$$

For $2 \leq k \leq n-1$ we have $A, B \geq d \geq 0$. The preceding row is log-concave with interval support, so all three gaps are nonnegative. At the two boundaries one of the neighboring output masses is zero, and there is nothing to check. We start from the point mass at one and induct. The recurrence also shows the masses are strictly positive on the support.

Next we identify the limit. For $t \geq 1$, join step $t+1$ to the earlier step X_i that it copies or reverses, and give this edge the sign $X_{t+1}X_i$. This builds a random recursive tree on the steps. The edge signs are independent of the tree and of each other, and each is $+1$ with probability $(1+a)/2$. Remove the edge from step 2 to step 1. This splits the steps into the part of step 1 and the part of step 2. A new step joins a part with probability equal to the share of that part among the steps so far. Hence the sizes of the two parts form a Pólya urn started from one ball of each color, and the share of the part of step 1 converges almost surely to a uniform variable U. Given the sequence of parts that the steps join, each part is a random recursive tree in its own time, with its own independent edge signs. So the sum of the steps in the part of step 1 is an elephant walk with a positive first step, observed at the size of that part. The sum in the part of step 2 is minus an independent copy of such a walk. Both sizes tend to infinity. By the almost sure convergence of S_k/k^a given $X_1 = 1$ [10, Theorem 3.7],

$$\frac{S_n}{n^a} \longrightarrow \mathcal{L}_a^{+-} := U^a \mathcal{L}_a' - (1-U)^a \mathcal{L}_a'' \qquad \text{almost surely,}$$

where $\mathcal{L}_a'$ and $\mathcal{L}_a''$ are copies of $\mathcal{L}_a$ and U, $\mathcal{L}_a'$, $\mathcal{L}_a''$ are independent. With two positive first steps the same split gives the limit $\mathcal{L}_a^{++} = U^a \mathcal{L}_a' + (1-U)^a \mathcal{L}_a''$.

These limits have entire densities. Let ϕ_a be the characteristic function of $\mathcal{L}_a$, and take $1 < r < 1/a$. Since $ra < 1$, we have $v^{ra} + (1-v)^{ra} \geq 1$ for $0 \leq v \leq 1$. Hence the

bound of Lemma D.1 gives

$$\left|\mathbb{E}e^{it\mathcal{L}_a^{+\pm}}\right| \le \int_0^1 |\phi_a(tv^a)|\,|\phi_a(t(1-v)^a)|\,dv \le 4e^{-c|t|^r}.$$

So, exactly as in the proof of that lemma, Fourier inversion gives densities f_a^{+-} and f_a^{++} that extend to entire functions. In particular they are smooth, so Lemma A.1 applies to both starts.

Now interpolate the lattice masses log-linearly. The result is log-concave. Its integral is within twice the largest atom of the step interpolation's. The largest atom goes to zero, because the count laws converge weakly to a law with a continuous density. Hence the limit is log-concave too. Symmetry follows from the representation of $\mathcal{L}_a^{+-}$. Since $h_a > 0$, also $f_a^{+-} > 0$. Now $(\log f_a^{+-})'$ is nonincreasing and is zero at zero. Suppose it were also zero at some $t > 0$. Then the density would be constant on $[0, t]$, which analyticity rules out. Hence $(f_a^{+-})' < 0$ for $t > 0$.

Next take two positive initial steps. We claim the first differences are nonnegative on the negative half-line, and use induction from the point mass at count two. Each entry on the negative half-line has both parents in the lower half of the previous row. The one exception is the central difference of an odd row, and there reflection dominance gives the sign. The derivative limit of Lemma A.1 then gives $(f_a^{++})'(t) \ge 0$ for $t < 0$. Finally $h_a = ((1+a)/2)f_a^{++} + ((1-a)/2)f_a^{+-}$. The second summand has strictly positive derivative there. □

Guérin, Laulin, Raschel, and Simon prove positivity of the conditional mode for memory sufficiently close to one [12, Remark 2.1(a)]. We need it for all $a \in (1/2, 1)$, in the stronger form of a positive slope at zero.

Lemma A.5 (the conditional mode is positive)**.** *For every $1/2 < a < 1$, we have $h_a'(0) > 0$. In particular the unique conditional mode lies strictly to the right of zero.*

Proof. The idea is to bound the reflection difference from below by something that survives rescaling. Write $f_n(k) = \mathbb{P}_+(A_n = k)$ and, for $z \equiv n \pmod 2$, $d_n(z) = f_n((n+z)/2) - f_n((n-z)/2)$. With a positive start, the reflection induction of Proposition A.3 is strict for $0 < z \le n$. Indeed $d_1(1) = 1$, and each noncentral entry of the upper half has a positive parent with a positive coefficient. The recurrence is

$$d_{n+1}(z) = \alpha_n(z)d_n(z+1) + \beta_n(z)d_n(z-1),$$
$$\alpha_n(z) = \frac{1}{2} - \frac{a(z+1)}{2n}, \qquad \beta_n(z) = \frac{1}{2} + \frac{a(z-1)}{2n}.$$

Fix $n_0 > 4a$. Divide by z and multiply by $\prod_{j=n_0}^{n-1}(1-2a/j)^{-1}$. Call the new values $v_n(z)$. They satisfy a backward averaging equation, with weights

$$q_+ = \frac{\alpha_n(z)(z+1)}{z(1-2a/n)}, \qquad q_- = \frac{\beta_n(z)(z-1)}{z(1-2a/n)}.$$

In the band we use below, these weights are nonnegative and add up to one. At $z = 1$ the parent 0 gets weight zero. Let z_n be the backward chain with these weights. Then

$$\mathbb{E}[z_n^2 \mid z_{n+1} = z] = \frac{n-4a}{n-2a}z^2 + \frac{3n-4a}{n-2a}.$$

With $\gamma_n = \Gamma(n-2a)/\Gamma(n-4a) \sim n^{2a}$, the process $Y_n = z_n^2/\gamma_n$ has backward drift $\delta_n = (3n-4a)/\{(n-2a)\gamma_n\}$. Note that $\sum_{n \ge n_0} \delta_n \to 0$ as $n_0 \to \infty$.

Fix $R > 0$. We start the chain at $z_N = 1$ if N is odd and at $z_N = 2$ if N is even. We stop it at row n_0, or when it leaves the band $0 < z/\sqrt{\gamma_n} \le R$. Since $\sqrt{\gamma_n} \sim n^a$ and $a < 1$, we

can take n_0 so large that $R\sqrt{\gamma_n}+1 \le n$ for all $n \ge n_0$. Then every state in the band and its two parents lie in the support, and $\alpha_n(z), \beta_n(z) \ge 0$ there. The chain uses the weights only at band states before it stops, so we need their nonnegativity only there. The chain moves by ± 1, and it can reach 0 only from $z = 1$, where the parent 0 has weight zero. So it cannot leave the band through the bottom, and on leaving the band $Y > R^2$. Up to the stopping time, the mean of Y is at most $z_N^2/\gamma_N + \sum_{n \ge n_0} \delta_n$. Taking n_0 and then N large, Markov's inequality shows that the chain leaves the band with probability below one half. Every stopped value of v is positive, by the strict reflection induction. The band states of row n_0 are the finitely many z with $0 < z \le R\sqrt{\gamma_{n_0}}$, and at row n_0 the normalizing product is empty, so $v_{n_0}(z) = d_{n_0}(z)/z$. Their minimum $m_0 > 0$ depends only on n_0 and R, not on N. The chain stops in this row with probability above one half. Since stopping is bounded, $v_N(z_N) \ge m_0/2$. The product normalizer is a positive constant times N^{2a} in the limit. Hence

$$\liminf_N N^{2a} \frac{d_N(z_N)}{z_N} > 0.$$

By Lemma A.1 for the conditional start, the first differences converge, and this limit is $4h_a'(0)$. So $h_a'(0) > 0$. Since h_a is unimodal, its mode is to the right of zero. □

B An analytic proof of the transition

Here we prove Theorem 3.2. The plan is to write the sign of the central curvature as a comparison between positive weights. We get the integral from the moment equation of Guérin, Laulin, Raschel, and Simon [12, (15)].

Put

$$\chi_k = 4^{-k}\binom{2k}{k}, \qquad H_a(y) = 1 - \sum_{k \ge 1} \frac{\chi_k}{2ak-1} y^k. \tag{B.1}$$

Note that H_a is strictly decreasing on $[0, 1)$, and $H_1(y) = \sqrt{1-y}$. When $a < 1$ each subtracted coefficient gets bigger, so $H_a(1) < 0$. (The series converges at $y = 1$. Its coefficients are at most a constant times $\chi_k/(2k-1)$, and these sum to one.) Hence H_a has exactly one root $Y_a \in (0, 1)$. Comparing coefficients, we also get

$$H_a(y) - 2ayH_a'(y) = (1-y)^{-1/2}. \tag{B.2}$$

We have $\partial_a H_a(y) > 0$ for $y > 0$ and $H_a'(Y_a) < 0$. By the implicit function theorem, Y_a is smooth and $Y_a' > 0$.

Lemma B.1 (a weighted formula for the central curvature)**.** *For* $1/2 < a < 1$,

$$g_a''(0) = -\frac{1}{\pi\Gamma(3-3a)} \int_0^{Y_a} y^{1/2} H_a(y)^{2-3a} \big[1 - (1+a)^2 y\big]\, dy. \tag{B.3}$$

The integral is absolutely convergent.

Proof. We multiply the limit by an independent exponential power to cancel the gamma factors in its moments. Let E be an independent exponential random variable of mean one and set $Z_a = E^a \mathcal{L}_a$. With the normalization of [12],

$$m_n = \frac{\Gamma(1+an)}{n!} \mathbb{E}\mathcal{L}_a^n, \qquad m_0 = m_1 = 1.$$

The bound $0 < m_n \le C_a^n$ follows from [13, Lemma 2.10 and Corollary 2.14]. Applied to the even moments, it gives $\mathbb{E}\exp(\delta|Z_a|) < \infty$ for small $\delta > 0$. Hence $M_a(z) = \mathbb{E}e^{zZ_a}$ is analytic on a strip about the imaginary axis. Compare its power series with the moment equation. We get

$$M_a(z) + azM_a'(z) = \frac{1+a}{2} M_a(z)^2 + \frac{1-a}{2} M_a(z) M_a(-z). \tag{B.4}$$

first near zero, and then on the whole strip by analytic continuation. Write $M_a(it) = A(t) + iB(t)$. Then

$$atA' = A^2 - A - aB^2, \qquad atB' = ((1+a)A - 1)B. \tag{B.5}$$

For $0 < y < Y_a$ define

$$t = \sqrt{y}\, H_a(y)^{-a}, \qquad A(t) = H_a(y)\sqrt{1-y}, \qquad B(t) = H_a(y)\sqrt{y}. \tag{B.6}$$

On the imaginary axis, this is the closed form of [12, Proposition 3.2], with their $G(it) = i\sqrt{y}$ and $(G(it)/(it))^{1/a} = H_a(y)$. We check the continuation directly from (B.5). As y runs over $(0, Y_a)$, t increases from zero to infinity, and by (B.2)

$$\frac{1}{t}\frac{dt}{dy} = \frac{1}{2yH_a(y)\sqrt{1-y}},$$

Substituting, (B.6) satisfies (B.5). Next, $H_a(0) = 1$, so the analytic map $y \mapsto yH_a(y)^{-2a}$ has derivative one at zero. From $t^2 = yH_a(y)^{-2a}$ we see that y is analytic in t^2 near zero. Hence A is even and analytic, and $B = tH_a(y)^{1+a}$ is odd and analytic, with $A(0) = 1$ and $B'(0) = 1$. These values fix both power series, because the coefficient recurrence has denominator $na - 1$ for even $n \geq 2$ and $na - a$ for odd $n \geq 3$, and both are positive. So near zero the parametrization equals $M_a(it)$. By uniqueness for the ODE (B.5) on $t > 0$, they agree for all $t > 0$.

Write $f(t) = \operatorname{Re} \mathbb{E} e^{it\mathcal{L}_a}$. Exponential randomization gives

$$A(t) = \int_0^\infty e^{-v} f(tv^a)\, dv.$$

We use the Fourier decay in Lemma D.1, which in particular makes every polynomial multiple of f integrable at infinity. For $0 < \operatorname{Re} s < 1/a$, absolute Fubini gives

$$J(s) := \int_0^\infty t^{s-1} A(t)\, dt = \Gamma(1-as)\mathcal{F}(s), \qquad \mathcal{F}(s) = \int_0^\infty t^{s-1} f(t)\, dt. \tag{B.7}$$

The function $\mathcal{F}$ is holomorphic on $\operatorname{Re} s > 0$. The integral defining J diverges at $s = 3$, the value needed for curvature. Two integrations by parts give a convergent expression there.

We send the root to zero before integrating by parts. Set $\theta = \arcsin\sqrt{y}$ and

$$u = F(\theta) = \sin^{-1/a}\theta\, H_a(\sin^2\theta), \qquad w(u) = a\cos\theta \sin^{1/a}\theta.$$

Here θ decreases from $\arcsin\sqrt{Y_a}$ to zero as u increases from zero to infinity. From (B.2),

$$F'(\theta) = -\frac{1}{a}\sin^{-1-1/a}\theta, \qquad t = u^{-a}.$$

Changing variables in J gives $J(s) = \int_0^\infty u^{-as} w(u)\, du$. Since $F'(\arcsin\sqrt{Y_a}) \neq 0$, the function w is smooth at zero. At infinity, $\theta \sim u^{-a}$, so its definition and the derivative formulas below give $w(u) = O(u^{-1})$, $w'(u) = O(u^{-2})$, and $w''(u) = O(u^{-3})$. For $0 < \operatorname{Re} s < 1/a$, the boundary terms therefore vanish twice, and (B.7) becomes

$$\mathcal{F}(s) = \frac{1}{\Gamma(3-as)} \int_0^\infty u^{2-as} w''(u)\, du. \tag{B.8}$$

The right side is holomorphic on $0 < \operatorname{Re} s < 3/a$ by the bounds at infinity and boundedness of w'' at zero. Thus (B.8) holds throughout this larger strip by the identity theorem, and $s = 3$ is permitted.

Differentiating through $u = F(\theta)$ gives

$$w'(u) = -a \sin^{2/a} \theta [1 - (1+a) \sin^2 \theta],$$
$$w''(u) = 2a \sin^{3/a} \theta \cos\theta [1 - (1+a)^2 \sin^2 \theta].$$

Insert the second expression in (B.8) at $s = 3$ and return to $y = \sin^2 \theta$. The powers of y reduce to $y^{1/2}$, giving the integral in (B.3). Fourier inversion supplies the factor $g_a''(0) = -\mathcal{F}(3)/\pi$. Finally, H_a has a simple zero at Y_a, and $2 - 3a > -1$, so the endpoint singularity is integrable. □

Proof of Theorem 3.2. Signs outside a short interval.

Let μ_a be the probability law on $(0, Y_a)$ with density proportional to $y^{1/2} H_a(y)^{2-3a}$, and put

$$\mathcal{R}(a) = (1+a)^2 \int y \, d\mu_a(y). \tag{B.9}$$

Lemma B.1 says that $g_a''(0)$ has the sign of $\mathcal{R}(a) - 1$.

When $a \le 2/3$, the extra weight H_a^{2-3a} decreases in y. By Chebyshev's association inequality, it lowers the mean of the density proportional to $y^{1/2}$, giving $\int y \, d\mu_a(y) \le 3Y_a/5$. At $a = 2/3$, the first three terms of (B.1) give

$$H_{2/3}(3/5) \le 1 - \frac{9}{10} - \frac{81}{1000} - \frac{9}{400} = -\frac{7}{2000} < 0.$$

Since Y_a increases, $Y_a \le Y_{2/3} < 3/5$ for $a \le 2/3$. Thus $\mathcal{R}(a) < (5/3)^2 (3/5)^2 = 1$.

When $a \ge 2/3$, the extra weight increases, giving a mean of at least $3Y_a/5$. We need the bound $Y_{2/3} > 117/200$. To prove it, note that χ_k decreases. At $y = 117/200$, the terms with $k \ge 4$ are at most

$$\sum_{k \ge 4} \frac{\chi_k y^k}{4k/3 - 1} \le \frac{\chi_4 y^4}{(13/3)(1-y)}.$$

Together with the first three terms this gives the exact bound

$$H_{2/3}(117/200) \ge \frac{929771}{135987200} > 0.$$

For every $a \ge 69/100$, therefore,

$$\mathcal{R}(a) > \left(\frac{169}{100}\right)^2 \frac{3}{5} \frac{117}{200} = \frac{10024911}{10000000} > 1.$$

Any zero must lie between $2/3$ and $69/100$.

A single crossing.

We now work on $I = [2/3, 69/100]$. To compare densities on a fixed interval, rescale by $y = Y_a u$, so u always lies in $(0, 1)$. We separate the endpoint zero by putting

$$w_k(a) = \frac{\chi_k Y_a^k}{2ak - 1}, \qquad K_a(u) = \frac{H_a(Y_a u)}{1-u} = \sum_{k \ge 1} w_k(a)(1 + u + \cdots + u^{k-1}). \tag{B.10}$$

Since $\sum w_k = 1$, we have $K_a \ge 1$ and $\partial_u K_a \ge 0$. Writing $s_a = 3a - 2 \in [0, 7/100]$, the rescaled measure has density proportional to

$$W_a(u) = u^{1/2} (1-u)^{-s_a} K_a(u)^{-s_a}. \tag{B.11}$$

To make the mean increase, we show that the score $\partial_a \log W_a(u)$ is strictly increasing in u and use the covariance identity for the derivative of an expectation.

First, $Y_a < 2/3$ on I. The first three terms give

$$H_{69/100}(2/3) \le -\frac{9767}{7087608} < 0,$$

and Y_a increases. So every series and derivative below converges uniformly on $I \times [0,1]$. The falling-factorial moments are

$$M_1 = \sum kw_k, \qquad M_2 = \sum k(k-1)w_k, \qquad M_3 = \sum k(k-1)(k-2)w_k.$$

Let $G(y) = (1-y)^{-1/2}$. Evaluate (B.2) and its first two derivatives at $Y = Y_a$. This gives

$$\begin{aligned} 2aM_1 &= G(Y), \\ 2aM_2 &= YG'(Y) - (2a-1)M_1, \\ 2a(M_3 + 2M_2) &= Y^2G''(Y) + M_2. \end{aligned} \tag{B.12}$$

In particular,

$$0 \le \partial_u K_a(u) \le \frac{M_2}{2} \le \frac{YG'(Y)}{4a} \le \frac{3\sqrt{3}}{8} < \frac{21}{32}. \tag{B.13}$$

For the memory derivatives of the kernel, put $d_k = 2k/(2ak-1)$. Differentiating $\sum w_k = 1$ gives

$$\frac{Y_a'}{Y_a} M_1 = \sum d_k w_k \le 6, \qquad \frac{Y_a'}{Y_a} \le \frac{12a}{G(Y)}. \tag{B.14}$$

Here d_k decreases in k and $d_1 \le 6$ on I. The logarithmic derivative $\partial_a \log w_k = kY_a'/Y_a - d_k$ strictly increases in k. Since $1 + u + \cdots + u^{k-1}$ is nondecreasing in k, the covariance identity for derivatives of expectations gives $\partial_a K_a(u) \ge 0$. The probability law here is formed by the w_k.

In $\partial_a \partial_u K_a$, discard the negative term $-d_k w_k$ from $\partial_a w_k$, then bound the derivative of $1 + u + \cdots + u^{k-1}$ by $k(k-1)/2$. Equations (B.12) and (B.14) give

$$\begin{aligned} \partial_a \partial_u K_a &\le \frac{Y_a'}{Y_a} \frac{M_3 + 2M_2}{2} \\ &\le 3\left(\frac{M_2}{G(Y)} + \frac{Y^2 G''(Y)}{G(Y)} \right) \le 3\left(\frac{3}{4} + 3 \right) = \frac{45}{4}. \end{aligned} \tag{B.15}$$

The last bounds use $M_2/G(Y) \le Y/[4a(1-Y)]$ and $Y^2G''(Y)/G(Y) = 3Y^2/[4(1-Y)^2]$. Since $K_a \ge 1$, $\partial_a K_a \ge 0$ and $\partial_u K_a \ge 0$, we obtain

$$\partial_a \left(\frac{\partial_u K_a}{K_a} \right) = \frac{\partial_a \partial_u K_a}{K_a} - \frac{\partial_u K_a \, \partial_a K_a}{K_a^2} \le \frac{45}{4}.$$

In (B.11), the endpoint term dominates the kernel terms, giving the strict bound

$$\begin{aligned} \partial_u \partial_a \log W_a(u) &= \frac{3}{1-u} - 3\frac{\partial_u K_a}{K_a} - s_a \partial_a \left(\frac{\partial_u K_a}{K_a} \right) \\ &\ge 3 - \frac{63}{32} - \frac{7}{100}\frac{45}{4} = \frac{39}{160} > 0. \end{aligned} \tag{B.16}$$

The score $\partial_a \log W_a$ is therefore strictly increasing in u. If ν_a denotes the normalized measure in (B.11), differentiation gives

$$\frac{d}{da} \int u \, d\nu_a(u) = \operatorname{Cov}_{\nu_a}\big(u, \partial_a \log W_a(u)\big) > 0.$$

This is the strict form of Chebyshev's association inequality. Indeed, the covariance is one half the expectation of $(U-V)(S(U)-S(V))$ for independent $U, V \sim \nu_a$ and the

strictly increasing score S. We may differentiate under the integral, because on this compact parameter interval the integrand is dominated by $C(1-u)^{-7/100}(1+|\log(1-u)|)$, which is integrable.

It follows that

$$\mathcal{R}(a) = (1+a)^2 Y_a \int u \, d\nu_a(u)$$

has strictly positive derivative in the interior of I. By the signs at the ends of I and the intermediate value theorem, $\mathcal{R}-1$ has exactly one zero in I. We already know it has none outside I. Now rewrite (B.3) as

$$g_a''(0) = \frac{\mathcal{Z}_a}{\pi\Gamma(3-3a)}\,[\mathcal{R}(a)-1], \qquad \mathcal{Z}_a = \int_0^{Y_a} y^{1/2} H_a(y)^{2-3a}\, dy > 0.$$

The same rescaling and domination show that the prefactor is continuously differentiable near the zero. At the zero $\mathcal{R}-1$ vanishes, so only $\mathcal{R}'$ contributes, and $\partial_a g_a''(0) = \mathcal{Z}_a\mathcal{R}'(a)/[\pi\Gamma(3-3a)] > 0$. By Theorem 3.1, g_a is then unimodal for $a < a_*$ and bimodal for $a > a_*$. For $a = a_*$ itself, let $a \uparrow a_*$. The inequality $g_a(x) \geq g_a(y)$ for $0 \leq x < y$ survives in the limit by joint continuity. So g_{a_*} is nonincreasing on $[0,\infty)$, and since it is not constant on any interval, its only mode is zero. □

With the same bounds on Y_a we can also get the sign of the fourth derivative.

Lemma B.2 (the quartic sign)**.** *For $2/3 \leq a \leq 69/100$, we have $g_a^{(4)}(0) < 0$. In particular, $B_* = -g_{a_*}^{(4)}(0) > 0$.*

Proof. We integrate by parts once more in the Mellin calculation. With u, w as in the proof of Lemma B.1, put

$$c_a = (1+a)^2, \qquad P_a(y) = \frac{3}{a}(1-y)(1-c_a y) - y(1-c_a y) - 2c_a y(1-y).$$

Since $dy/du = -2a y^{1+1/(2a)}\sqrt{1-y}$, differentiation gives $w'''(u) = -2a^2 y^{2/a} P_a(y)$. This derivative is bounded at zero and is $O(u^{-4})$ at infinity. The boundary terms vanish for $0 < \operatorname{Re} s < 3/a$, so (B.8) gives

$$\mathcal{F}(s) = -\frac{1}{\Gamma(4-as)} \int_0^\infty u^{3-as} w'''(u)\, du.$$

The right side is holomorphic for $0 < \operatorname{Re} s < 4/a$. For $1/2 < a < 4/5$, we may therefore set $s = 5$ and use $g_a^{(4)}(0) = \mathcal{F}(5)/\pi$ to obtain

$$g_a^{(4)}(0) = \frac{a}{\pi\Gamma(4-5a)} \int_0^{Y_a} y^{3/2}(1-y)^{-1/2} H_a(y)^{3-5a} P_a(y)\, dy. \tag{B.17}$$

The integral converges absolutely because H_a has a simple zero and $3-5a > -1$.

Now let $a \in [2/3, 69/100]$. The root bounds in the preceding proof give $1/2 < Y_a < 2/3$. For the coefficient comparison only, we use the larger interval $2/3 \leq a \leq 7/10$. Expanding P_a gives, for $y \geq 0$,

$$\begin{aligned} P_a(y) &= \frac{3}{a} - \left(2a^2 + 7a + 9 + \frac{6}{a}\right) y + \left(3a^2 + 9a + 9 + \frac{3}{a}\right) y^2 \\ &\leq \frac{9}{2} - 23y + 22y^2 =: \widehat{P}(y). \end{aligned}$$

Here termwise lower bounds give $8/9 + 14/3 + 9 + 60/7 = 1457/63 > 23$ for the coefficient of $-y$, and termwise upper bounds give $147/100 + 63/10 + 9 + 9/2 = 2127/100 < 22$ for

that of y^2. Since $\widehat{P}(0) > 0$, $\widehat{P}(1/2) = -3/2$, and $\widehat{P}(2/3) = -19/18$, the quadratic $\widehat{P}$ has exactly one root y_0 in $(0, Y_a)$, with $\widehat{P} > 0$ before it and $\widehat{P} < 0$ after it. Integrating directly, for $1/2 \le Y \le 2/3$,

$$Y^{-5/2} \int_0^Y y^{3/2} \widehat{P}(y)\, dy = \frac{9}{5} - \frac{46}{7} Y + \frac{44}{9} Y^2 \le -\frac{83}{315} < 0.$$

The middle expression is convex in Y, so it suffices to check $Y = 1/2$ and $Y = 2/3$.

The weight $\omega_a(y) = (1-y)^{-1/2} H_a(y)^{3-5a}$ is what remains, and it is positive and increasing on $(0, Y_a)$. Since $\widehat{P}$ swaps signs only at y_0, we have $\omega_a(y)\widehat{P}(y) \le \omega_a(y_0)\widehat{P}(y)$ on the whole interval. Hence

$$\int_0^{Y_a} y^{3/2} \omega_a(y) P_a(y)\, dy \le \omega_a(y_0) \int_0^{Y_a} y^{3/2} \widehat{P}(y)\, dy < 0.$$

On $[2/3, 69/100]$ the prefactor in (B.17) is positive, and the lemma follows. □

C Signs at the first lattice point

We prove Lemma 2.3. Put $\xi_t = P_t(1)/P_t(0)$. By (2.1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1 - \varepsilon - \frac{a}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{a\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/a$ for $t \ge 2$. For $t = 2$ this holds because $a < 1 - \varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \le \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{a}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \ge 2$, and $\xi_t < 1$ whenever $t \le a/\varepsilon$.

Now (2.2) with $k = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \ge 2$

$$s_{t+1} = \big(1 - r_t(2)\big) s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{a}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \ge 1 - \varepsilon - a = \varepsilon > 0$. If $t < a/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > a/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ξ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. To remove the positive multiplier, let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \ge \frac{1}{3}$, then $a/\varepsilon \le 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-, +)$. This proves the claim. □

D Regularity of the limit densities

We prove the regularity used in Sections 3 and A.

Lemma D.1 (entire densities and parameter continuity)**.** *For every $a \in (\frac{1}{2}, 1)$ and $1 < r < 1/a$, the characteristic function ϕ_a of $\mathcal{L}_a$ satisfies*

$$|\phi_a(t)| \le 2e^{-c|t|^r} \qquad (t \in \mathbb{R})$$

for some $c = c(a, r) > 0$. The same c works on a sufficiently small neighborhood of any fixed a. For every compact $J \subset (1/2, 1)$ and every $\beta > 0$, there is $C_{J,\beta} < \infty$ such that

$$|\partial_t \phi_a(t)| + |\partial_a \phi_a(t)| \le C_{J,\beta}(1 + |t|)^{-\beta} \qquad (a \in J,\ t \in \mathbb{R}).$$

For every integer $k \ge 0$, the derivatives $\partial_u^k h_a(u)$ and $\partial_a \partial_u^k h_a(u)$ exist and are jointly continuous in (a, u). For every $q \in [0, 1]$, the density of the limit with first-step bias q extends to an entire function and is not constant on an interval of positive length.

Proof. We first check weak continuity in a to obtain a Fourier bound local in memory. On a compact $J \subset (1/2, 1)$, the martingale $\mathcal{M}_n(a) = \Gamma(n)S_n/\Gamma(n + a)$ of [10, (2.4), Appendix A] satisfies

$$\sup_{a \in J} \mathbb{E}|\mathcal{L}_a - \mathcal{M}_n(a)|^2 \le C_J \sum_{k \ge n} k^{-2 \min J} \longrightarrow 0.$$

The walk's centered next step has conditional variance at most one, and the gamma ratios are uniformly bounded on J. The finite laws are continuous in a, so $\phi_a(t)$ is jointly continuous, uniformly on compact frequency intervals.

The fixed-point equation of [13, Theorem 1.1 and (2.10)] gives

$$|\phi_a(t)| \le \int_0^1 |\phi_a(tv^a)|\, |\phi_a(t(1 - v)^a)|\, dv.$$

Here ϕ_a is the characteristic function conditional on a positive first step. The random sign disappears after taking absolute values, since $|\phi_a(-s)| = |\phi_a(s)|$. Fix a_0 and $1 < r < 1/a_0$, and choose J about a_0 with $r \max J < 1$. Put $D(v) = v^{r \max J} + (1 - v)^{r \max J} - 1$. We have $D(v) > 0$ for $0 < v < 1$. Dominated convergence therefore permits $K > 1$ such that

$$R := 2 \int_0^1 e^{-K^r (\log 2) D(v)}\, dv < 1.$$

The law of $\mathcal{L}_{a_0}$ has a density [13, Theorem 1.3], so the Riemann–Lebesgue lemma gives $T > 0$ with $|\phi_{a_0}(t)| < 2^{-K^r}$ for $|t| \ge T$. Joint continuity, after shrinking J, gives $|\phi_a(t)| \le 2^{1-K^r}$ for $a \in J$ and $T \le |t| \le KT$. Set $c = (\log 2)/T^r$ and $b(t) = 2e^{-ct^r}$ for $t \ge 0$. On $[0, T]$ we have $b \ge 1$, and on $[T, KT]$ we have $b \ge 2^{1-K^r}$. Thus $|\phi_a| \le b$ on $[0, KT]$.

For $t \ge KT$,

$$\frac{\int_0^1 b(tv^a) b(t(1 - v)^a)\, dv}{b(t)} \le 2 \int_0^1 e^{-ct^r D(v)}\, dv \le R < 1.$$

Suppose the bound $|\phi_a| \le b$ stops at a finite $t_* \ge KT$. By continuity it still holds on $[0, t_*]$. But at t_* the fixed-point inequality gives $|\phi_a(t_*)| \le Rb(t_*) < b(t_*)$, which is strict. By continuity the bound holds a little past t_*, a contradiction. Hence it holds on $[0, \infty)$, and on $\mathbb{R}$ by evenness.

Since $r > 1$, the Fourier integral converges absolutely and locally uniformly for complex arguments. Indeed, for every $B < \infty$, $\int_{\mathbb{R}} e^{B|t|} |\phi_a(t)|\, dt < \infty$. It therefore defines an entire function. After any fixed number of spatial derivatives, the bound on J still supplies an integrable majorant, giving joint continuity by dominated convergence. For initial bias q, the triangle inequality applied to $q\phi_a(t) + (1 - q)\phi_a(-t)$ gives the same bound. A density of integral one cannot be constant on $\mathbb{R}$. If it were constant on an interval, the identity theorem would make it constant everywhere.

To differentiate in memory, we first work on bounded frequency intervals. The moment recursion of [13, Theorem 1.4] is

$$m_0 = m_1 = 1, \qquad m_n(z) = \frac{\sum_{j=1}^{n-1} \theta_j(z) m_j(z) m_{n-j}(z)}{nz - \theta_n(z)}, \qquad \theta_j(z) = \begin{cases} 1 & j \text{ even}, \\ z & j \text{ odd}. \end{cases}$$

Choose a closed complex disc about a_0 on which $\operatorname{Re} z \geq \alpha > 1/2$, $|z| < 1$ and $|\operatorname{Im} z| \leq \rho$. All denominators for $n \geq 2$ have real part at least $n\alpha - 1 > 0$. Put $A = 2\alpha - 1$ and $B = A^{-1}$. Induction gives

$$|m_n(z)| \leq AB^n \quad (n \geq 1), \qquad \frac{(n-1)A^2}{n\alpha - 1} \leq A \quad (n \geq 2).$$

For $x > 0$ and real y, the gamma product gives

$$\left| \frac{\Gamma(x)}{\Gamma(x+iy)} \right|^2 = \prod_{j \geq 0} \left(1 + \frac{y^2}{(x+j)^2} \right) \leq \exp\left(\frac{y^2}{x^2} + \frac{y^2}{x} \right).$$

Taking $x = 1 + n \operatorname{Re} z$ and $y = n \operatorname{Im} z$ gives

$$\left| \frac{m_n(z)}{\Gamma(1+nz)} \right| \leq \frac{CB_0^n}{\Gamma(1+n\alpha)}.$$

Here C, B_0 are independent of n, z. We used monotonicity of Γ on $[2, \infty)$ for $n \geq 2$, and enlarged C for $n = 0, 1$. Thus $\sum_{n \geq 0} m_n(z)(it)^n / \Gamma(1+nz)$ converges locally uniformly in complex t and z in the disc's interior. For real $z = a$ this is $\phi_a(t)$, by the moment formula and exponential moments of [13, Corollary 2.11]. Cauchy's formula gives local joint continuity of every frequency derivative and its memory derivative.

We now obtain bounds at large frequencies without a numerical input. Shrink J about a_0 so that it lies inside this disc and retains the uniform exponential bound. For $t \geq 0$ write $f_a(t) = |\phi_a(t)|$, $u_a(t) = |\partial_t \phi_a(t)|$ and $w_a(t) = |\partial_a \phi_a(t)|$. The smoothing equation is

$$\phi_a(t) = \int_0^1 \phi_a(tv^a) \{ p\phi_a(t(1-v)^a) + (1-p)\phi_a(-t(1-v)^a) \} \, dv, \qquad p = (1+a)/2.$$

The local bounds just proved justify differentiation under this integral, since $v^{\min J} |\log v|$ is integrable. Put $x = tv^a$, $y = t(1-v)^a$ and, for nonnegative z, define

$$(\mathcal{K}_a z)(t) = \int_0^1 \{ z(x) f_a(y) + f_a(x) z(y) \} \, dv.$$

Taking absolute values and discarding factors $v^a, (1-v)^a \leq 1$ in the frequency derivative gives

$$u_a \leq \mathcal{K}_a u_a, \qquad w_a \leq \mathcal{K}_a w_a + S_a,$$

where

$$S_a(t) = \int_0^1 \{ x |\log v| u_a(x) f_a(y) + y |\log(1-v)| f_a(x) u_a(y) + f_a(x) f_a(y) \} \, dv.$$

The last term bounds the derivative of p, since $p' = 1/2$ and $|\phi_a(y) - \phi_a(-y)| \leq 2 f_a(y)$.

Fix $\beta > 0$. Splitting at $1/2$ and reflecting the second half gives

$$K_{a,\beta}(t) = 2 \int_0^{1/2} \left\{ \left(\frac{1+t}{1+x} \right)^\beta f_a(y) + \left(\frac{1+t}{1+y} \right)^\beta f_a(x) \right\} dv \longrightarrow 0$$

uniformly for $a \in J$ as $t \to \infty$. Indeed, $y \geq t/2$ bounds the first term by $2(1+t)^\beta e^{-c(t/2)^r}$ and the second ratio by 2^β. Also $x \geq tv^{\max J}$, so the exponential bound and dominated convergence make the second integral tend to zero uniformly. Choose T so that $K_{a,\beta}(t) \leq 1/2$ for all $a \in J$, $t \geq T$. For $X \geq T$ let

$$U_X = \max_{a \in J,\, 0 \leq t \leq X} (1+t)^\beta u_a(t).$$

All arguments x, y are at most t. The first smoothing inequality gives

$$U_X \le \max\{U_T, U_X/2\}, \qquad \text{hence } U_X \le U_T.$$

The bound is independent of X. This proves arbitrary polynomial decay for u_a.

The same decay holds for S_a. On $0 < v \le 1/2$, the first logarithmic term uses the exponential decay of $f_a(y)$ and boundedness of $u_a(x)$, with $x|\log v| \le tv^{\min J}|\log v|$. The second uses arbitrary polynomial decay of $u_a(y)$, $f_a(x) \le 1$, and $|\log(1-v)| \le \log 2$. The last uses the exponential decay of $f_a(y)$. Reflection covers the other half. Thus $S_\beta := \sup_{a\in J, t\ge 0}(1+t)^\beta S_a(t) < \infty$. Defining W_X as above with w_a in place of u_a, we obtain

$$W_X \le \max\{W_T, W_X/2 + S_\beta\} \quad \Longrightarrow \quad W_X \le \max\{W_T, 2S_\beta\}.$$

Conjugate symmetry extends both derivative bounds to negative t. A finite cover gives the stated bounds on every compact memory interval. Since β is arbitrary, multiplication by any fixed power of t leaves an integrable uniform majorant. Differentiation under Fourier inversion and dominated convergence prove the claimed joint continuity of $\partial_a \partial_u^k h_a$ for every k. □

Statement on AI use

I used two AI tools for this paper, with their contributions described here. Claude Opus 5.5 (Anthropic, through Claude Code) ran literature searches, citation checks, and referee-style reviews, checked computations in the threshold and quartic-top proofs, and made final copy edits and link updates. The reviews informed the framing and the shortening of an earlier combined manuscript, and suggested proving without computation that the bias boundary is positive, that the transition is unique, and that the critical top is quartic. All proofs in the paper were written by me, and the aforementioned suggestions are the only Claude suggestions that led to a new proof or result. The Lean files and their coverage map in lean/PaperD were generated with Claude Opus 5.5 from my statements and proofs; I checked that they build and audited their axioms. GPT-6 Astra (OpenAI, through Codex) edited my drafts for grammar, concision, and consistent notation, checked the citations, and read the proofs for errors. It reported inconsistent notation, and it helped me close a gap, after the setup, in my first proof that $b_a > 0$. Neither tool wrote a section of the paper. I read every cited source in full except Eisenberger (1964) and Robertson and Fryer (1969), which are cited only for results stated in their abstracts. Section 4 separates the written proofs, the formal proofs with their named classical hypotheses, and the numerical certificates. I take full responsibility for the paper, including its proofs and computations.

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